

Data Quality & Forensics Report

Purpose

Assess whether the recovery trend can be trusted and identify data defects that could materially change business conclusions.

Core principles

1. Raw files are immutable.
2. Account ID is the operational anchor.
3. Exact duplicates are removed programmatically.
4. Conflicting records are quarantined rather than silently overwritten.
5. Observational associations are not labeled causal.

Findings

- `calls_borrower_mismatch_vs_account`: 89,939. Event `borrower_id` differs from account master `borrower_id`.
- `whatsapp_events_borrower_mismatch_vs_account`: 59,673. Event `borrower_id` differs from account master `borrower_id`.
- `account_status_history_borrower_mismatch_vs_account`: 59,035. Event `borrower_id` differs from account master `borrower_id`.
- `targeting_outside_campaign_window`: 36,333. Target date outside campaign definition window.
- `status_recorded_before_event`: 30,191. `recorded_at` earlier than `event_at`.
- `payments_borrower_mismatch_vs_account`: 25,113. Event `borrower_id` differs from account master `borrower_id`.
- `latest_history_status_differs_from_account`: 22,295. Latest status history differs from account master status.
- `accounts_borrower_not_in_borrowers`: 2,913. Master borrower IDs absent from borrowers table.
- `calls_duplicate_id_groups`: 1,350. IDs appearing more than once.
- `calls_duplicate_id_extra_rows`: 1,350. Rows beyond first per duplicate ID.
- `employee_codes_with_multiple_agent_ids`: 1,099. Employee codes associated with multiple agent IDs.
- `agent_ids_with_multiple_employee_codes`: 1,000. Agent IDs associated with multiple employee codes.
- `whatsapp_events_duplicate_id_groups`: 600. IDs appearing more than once.
- `whatsapp_events_duplicate_id_extra_rows`: 600. Rows beyond first per duplicate ID.
- `payments_duplicate_id_groups`: 500. IDs appearing more than once.
- `payments_duplicate_id_extra_rows`: 500. Rows beyond first per duplicate ID.
- `accounts_missing_borrower`: 455. Accounts without `borrower_id` in master.
- `calls_conflicting_duplicate_ids`: 11. Duplicate call IDs with at least one conflicting field.
- `borrower_mismatch_payments`: 0. See derived master-borrower consistency test.

Golden-data treatment

- `golden\_payments.csv`: exact duplicate rows removed; reused payment references flagged.
- `golden\_calls.csv`: exact duplicate rows removed; conflicting call IDs quarantined.
- `golden\_account\_month.csv`: account-month analytical spine anchored to account master; event borrower IDs replaced by account master borrower ID where available.
- August 2026 is marked partial.

Limitations

The dataset does not contain a time-varying balance field, so the recovery KPI is a proxy rather than a true opening-balance recovery rate. Campaign exposure windows are inconsistent, and agent identities are unstable. These limitations prevent a high-confidence causal claim.